

MENTAL HEALTH MOOD TRACKER

An AI-Powered Emotion Detection Web Application
Using Natural Language Processing & Machine Learning

Field	Details
Name	Adithya R
Course	Bachelor of Computer Applications (BCA)
Specialization	Artificial Intelligence & Machine Learning
Year	3rd Year — 2026
Project Type	Resume / Portfolio Project
Dataset	Emotions Dataset for NLP (Kaggle) — 20,000 sentences
Model Accuracy	Logistic Regression: 87.48% BERT: 93.77%

■ Live App: mental-health-mood-tracker-ai.streamlit.app ■ GitHub:
github.com/Adithyar-15/-Mental-Health-Mood-Tracker

1. Abstract

Mental health awareness has become increasingly important in today's fast-paced world. Many individuals struggle to recognize and track their own emotional patterns over time. This project presents an AI-powered Mental Health Mood Tracker — a web application that uses Natural Language Processing (NLP) and Machine Learning (ML) to automatically detect emotions from journal entries written by users.

The application classifies text into six emotions: Joy, Sadness, Anger, Fear, Love, and Surprise. It maintains mood history and displays interactive charts to help users understand emotional patterns. The system achieved 87.48% accuracy using Logistic Regression with TF-IDF vectorization, and 93.77% accuracy using a fine-tuned BERT transformer model.

2. Problem Statement

Mental health issues often go undetected because people fail to recognize emotional patterns in their daily lives. Traditional mood tracking requires manual analysis with no objective insights.

- People are unaware of their recurring emotional patterns
- Manual diary analysis is time-consuming and subjective
- Early warning signs of mental health issues go unnoticed
- No easy tool exists for objective, AI-powered emotion tracking

3. Objectives

- Build an NLP pipeline to preprocess and clean text data
- Train a Machine Learning model to classify emotions from text with high accuracy
- Fine-tune a BERT transformer model for improved performance
- Develop a user-friendly web interface using Streamlit
- Store journal entries and display mood trend visualizations
- Deploy the application live on the internet with a shareable link

4. Technologies Used

Technology	Purpose	Version
Python	Core programming language	3.12
Pandas	Data loading and manipulation	3.0.3
NLTK	Natural Language Processing	3.9.4
Scikit-learn	ML model training & evaluation	1.8.0

TF-IDF Vectorizer	Text to numerical conversion	Sklearn
Logistic Regression	Emotion classification model	Sklearn
HuggingFace Transformers	BERT model fine-tuning	Latest
PyTorch	Deep learning framework	Latest
Streamlit	Web application framework	1.58.0
Plotly	Interactive data visualization	6.7.0
SQLite	Journal entries database	Built-in
GitHub	Version control & code hosting	-
Streamlit Cloud	Live deployment platform	-

5. Dataset Description

Attribute	Details
Dataset Name	Emotions Dataset for NLP
Source	Kaggle — by Praveengovi
Total Records	20,000 sentences
Number of Emotion Classes	6 (Joy, Sadness, Anger, Fear, Love, Surprise)
Train Split	16,000 samples (80%)
Test Split	4,000 samples (20%)

5.1 Emotion Distribution

Emotion	Sample Count	Percentage	Difficulty
Joy	6,761	33.8%	Easy (most data)
Sadness	5,797	29.0%	Easy
Anger	2,709	13.5%	Medium
Fear	2,373	11.9%	Medium
Love	1,641	8.2%	Hard
Surprise	719	3.6%	Hard (least data)

6. Methodology

6.1 Text Preprocessing Pipeline

Step	Process	Example
1	Lowercasing	HAPPY → happy
2	Remove special characters	happy!!! → happy
3	Remove stopwords	i am feeling → feeling
4	Lemmatization	running → run, feeling → feeling

6.2 Feature Extraction — TF-IDF

Text was converted to 5,000-dimensional numerical vectors using TF-IDF Vectorization. TF-IDF assigns higher scores to rare, meaningful words and lower scores to common words, making it highly effective for emotion classification.

6.3 Model Training — Logistic Regression

- Excellent compatibility with high-dimensional TF-IDF features
- Fast training time (under 30 seconds on CPU)
- Strong baseline accuracy of 87.48% on test data
- Industry-standard approach for text classification tasks

6.4 BERT Fine-tuning

Parameter	Value
Base Model	bert-base-uncased (Google)
Epochs	3
Batch Size	16
Learning Rate	2e-5 (AdamW optimizer)
Max Sequence Length	128 tokens
Hardware	NVIDIA T4 GPU (Google Colab)
Training Time	~20 minutes

7. Results & Performance

7.1 Model Comparison

Model	Accuracy	Training Time	Status
Logistic Regression + TF-IDF	87.48%	~30 seconds	Deployed Live
Fine-tuned BERT	93.77%	~20 minutes	Research Only

7.2 Per-Emotion Performance

Emotion	LR Accuracy	BERT Accuracy	Improvement
Joy	90%	95%	+5%
Sadness	92%	97%	+5%
Anger	86%	94%	+8%
Fear	82%	90%	+8%
Love	77%	86%	+9%
Surprise	65%	79%	+14%

8. Web Application Features

- Real-time emotion detection from journal entry text
- Confidence score displayed with animated progress bar
- Animated emotion bubble indicators (Joy, Love, Sad, Fear, Surprise, Anger)
- Automatic saving of entries to SQLite database with timestamp
- Mood trend bar chart showing emotions over time using Plotly
- Emotion breakdown donut pie chart
- Dashboard with total entries, top emotion, and average confidence stats
- Recent journal entries table with date, emotion and confidence
- Beautiful responsive UI with custom CSS — soft pink aesthetic design
- Deployed live on Streamlit Cloud — accessible from any device worldwide

9. Challenges & Solutions

Challenge	Solution
Imbalanced dataset (Surprise: 719 vs Joy: 6761)	Documented as limitation; future fix using class weights
Stemmer producing non-real words (happi, believ)	Switched to WordNet Lemmatizer for real English words
BERT model too large to deploy (418MB)	Deployed LR model; documented BERT as research experiment
Git not recognized in VS Code terminal	Installed Git for Windows and configured system PATH
GitHub authentication popup not clickable	Used Personal Access Token (PAT) for secure authentication

10. Future Scope

- Deploy BERT model using Hugging Face Spaces for higher accuracy in production
- Add multi-label emotion detection (detecting multiple emotions simultaneously)
- Implement user authentication for personalized mood tracking across devices
- Add weekly mood summary reports sent via email notifications
- Expand emotion categories beyond 6 classes (e.g., disgust, anticipation)
- Build mobile application version using Flutter or React Native
- Add therapist dashboard for professional mental health monitoring
- Integrate voice input for hands-free journal entry

11. Conclusion

This project successfully demonstrates the application of Natural Language Processing and Machine Learning in the domain of mental health technology. Starting from raw data collection to model training, web application development, and live deployment — the complete ML pipeline was implemented and deployed within 5 days.

The Logistic Regression model achieved 87.48% accuracy, while the BERT model demonstrated that transformer-based approaches can push accuracy to 93.77%. The live web application provides a practical, user-friendly interface for emotion tracking with real-world applicability.

This project showcases skills in Python, NLP, Machine Learning, Deep Learning (BERT), Web Development (Streamlit), Data Visualization (Plotly), Database Management (SQLite), Version Control (GitHub), and Cloud Deployment — making it a comprehensive portfolio piece for placement interviews in AI/ML roles.

References

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